Determinants

2x2 determinants

1 Find $\det \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$.

3x3 determinants via cofactor expansion

2 Find $\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$ by cofactor expansion along the first row.

The determinant product rule

3 For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, verify $\det(AB) = \det(A)\det(B)$.

The invertibility criterion

4 State the invertibility criterion in terms of the determinant.

5 Is $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ invertible? Justify using the determinant.

Applying the invertibility criterion

6 For what value(s) of k is $\begin{bmatrix} k & 2 \\ 3 & k \end{bmatrix}$ NOT invertible?

More 2x2 determinant practice

7 Find $\det \begin{bmatrix} 7 & 2 \\ 3 & 5 \end{bmatrix}$.

More 3x3 cofactor expansion practice

8 Find $\det \begin{bmatrix} 2 & 0 & 1 \\ 3 & 1 & 2 \\ 1 & 0 & 4 \end{bmatrix}$ by expansion along the second column (it has two zeros).

Determinants of triangular matrices

9 State the shortcut for the determinant of a triangular matrix (all entries above, or below, the main diagonal are zero).

10 Use this shortcut to find $\det \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 1 & 7 & 4 \end{bmatrix}$.

Determinants and row operations

11 If you swap two rows of a matrix, how does the determinant change?

12 If a matrix has two identical rows, what must its determinant be, and why?

Application: area of a parallelogram

13 The area of the parallelogram spanned by $\mathbf{u} = \langle a, b\rangle$ and $\mathbf{v} = \langle c, d\rangle$ equals $\left|\det \begin{bmatrix} a & b \\ c & d \end{bmatrix}\right|$. Find the area for $\mathbf{u} = \langle 3, 1\rangle$, $\mathbf{v} = \langle 1, 4\rangle$.

Scalar multiples and the determinant

14 For an $n \times n$ matrix A, $\det(cA) = c^n \det(A)$. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = 2A$, find $\det(B)$ using this rule, then verify directly.

The determinant of the transpose

15 Verify that $\det(A^T) = \det(A)$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

The determinant of the inverse

16 For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, verify that $\det(A^{-1}) = \dfrac{1}{\det A}$.

A quick mental-math determinant

17 Find $\det \begin{bmatrix} -2 & 5 \\ 0 & 3 \end{bmatrix}$ (notice it's already triangular).

18 Find $\det \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix}$, and note what it tells you about invertibility.